Brain Tasks DevOps Project

1. Project Overview

This project demonstrates a complete DevOps pipeline by deploying a React-based application on AWS using Docker, Amazon ECR, AWS CodeBuild, and Amazon EKS.

2. Architecture

GitHub → CodeBuild → Docker → Amazon ECR → Amazon EKS → LoadBalancer

3. Tools Used

GitHub, Docker, AWS CodeBuild, Amazon ECR, Amazon EKS, kubectl, eksctl, CloudWatch

4. Application Setup

Cloned application from GitHub and verified it runs on port 3000 using 'serve'.

5. Dockerization

Created Dockerfile using AWS public ECR nginx image to avoid Docker Hub rate limits.

6. ECR Setup

Created ECR repository and pushed Docker image using AWS CLI.

7. EKS Cluster Setup

Created EKS cluster using eksctl and configured kubectl to interact with cluster.

8. Kubernetes Deployment

Deployed application using deployment.yaml and service.yaml. Service type used: LoadBalancer.

9. CI/CD Pipeline

Configured AWS CodeBuild to automate Docker build and push to ECR.

10. Monitoring

Used AWS CloudWatch logs to monitor build and deployment.

11. Troubleshooting

Resolved issues like GitHub connection errors, IAM permission issues, Docker Hub rate limits, and ECR repository errors.

12. Final Output

Application URL: http://a70808154209647d79fbbc58b8d6935e-163265015.us-east-1.elb.amazonaws.com/

▭ STEP 1: Launch EC2 (Your Main Work Machine)

👉 Create instance in Amazon EC2

Recommended:

- OS: Amazon Linux 2
- Instance: t3.medium
- Storage: 20GB

Security Group:

- Port 22 (SSH)
- Port 3000 (App testing)
- Port 80 (optional)

☐ STEP 2: Install All Required Tools

sudo yum update -y

Git

sudo yum install git -y

NodeJS

curl -fsSL https://rpm.nodesource.com/setup_18.x | sudo bash -

sudo yum install -y nodejs

Docker

sudo yum install docker -y

sudo systemctl start docker

sudo systemctl enable docker

sudo usermod -aG docker ec2-user

👉 Logout & login again after this step.

Verify: docker -version

```
[ec2-user@ip-172-31-42-225 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-42-225 ~]$
```

AWS CLI

sudo yum install aws-cli -y

kubectl

curl -o kubectl https://s3.us-west-2.amazonaws.com/amazon-eks/1.27.0/2023-11-14/bin/linux/amd64/kubectl

chmod +x kubectl

sudo mv kubectl /usr/local/bin/

eksctl

curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp

sudo mv /tmp/eksctl /usr/local/bin

🚀 Step 3. Clone & Run Application (Port 3000)

git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
cd Brain-Tasks-App

```
[ec2-user@ip-172-31-42-225 ~]$ git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 10 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 20.38 MiB/s, done.
[ec2-user@ip-172-31-42-225 ~]$ cd Brain-Tasks-App
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ ls
README.md  dist
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$
```

Install serve

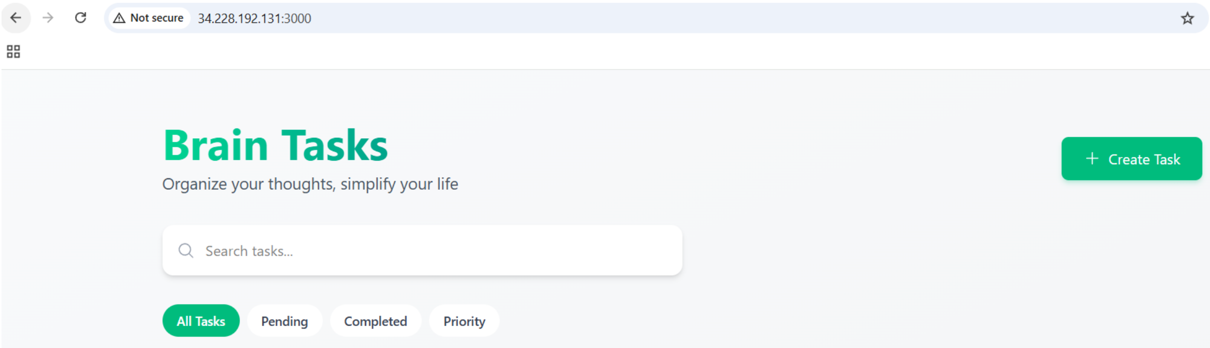
npm install -g serve

Run app

serve -s dist -l 3000

👉 Verify:

http://34.228.192.131:3000



STEP 4: Dockerize App

Create Dockerfile – vi Dockerfile

FROM public.ecr.aws/nginx/nginx:alpine

COPY dist/ /usr/share/nginx/html/

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

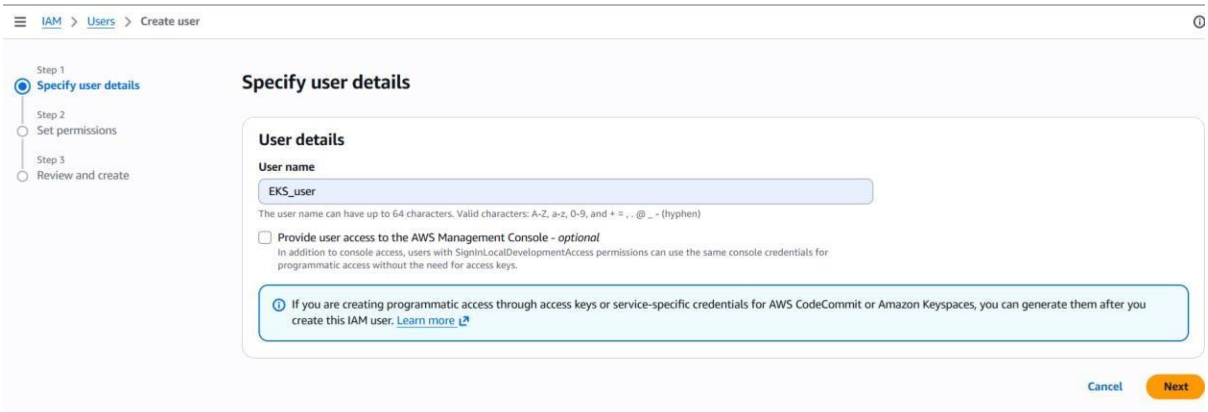
Build & Run

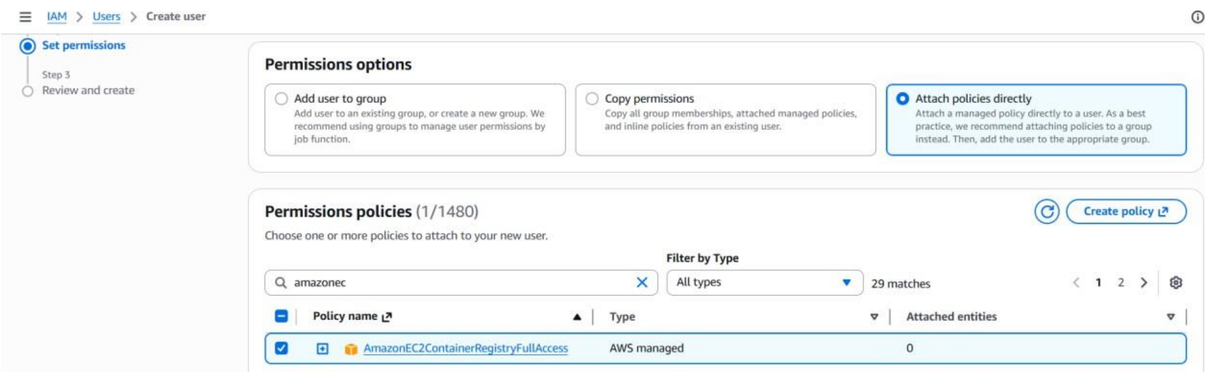
docker build -t brain-tasks-app .

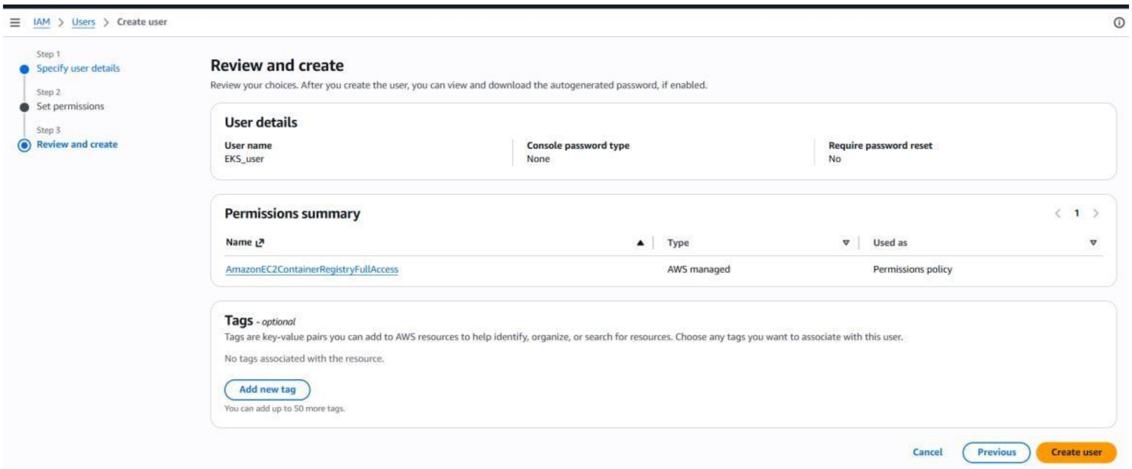
docker run -d -p 3000:80 brain-tasks-app

STEP 5: Push to ECR

To Create a IAM user








```
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ aws configure
AWS Access Key ID [None]: AKIAZUJYQDXNDM4DL3NW
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$
```

aws ecr create-repository --repository-name brain-tasks-app aws ecr get-login-password --region us-east-1 | \

docker login --username AWS --password-stdin 662080331226.dkr.ecr.us-east-1.amazonaws.com

docker tag brain-tasks-app 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest

docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest

```
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 662080331226.dkr.e
cr.us-east-1.amazonaws.com
WARNING! Your password will be stored unencrypted in /home/ec2-user/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ docker images
REPOSITORY        TAG       IMAGE ID       CREATED         SIZE
brain-tasks-app   latest    5ce86268199b   9 minutes ago   62.5MB
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ docker tag brain-tasks-app:latest 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
eba317107e43: Pushed
36c003a300d1: Pushed
2b12ccfb247a: Pushed
646d544dc619: Pushed
f7b74afe8798: Pushed
00a43f630638: Pushed
f4eddaa4c303: Pushed
e6082cdc2516: Pushed
29df493baa13: Pushed
latest: digest: sha256:41f43c3e097729c302658e08e661b098c32069a123a69ce37ac13a66b3d35eb3 size: 2199
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$
```

☸ STEP 6: Create EKS Cluster

eksctl create cluster \

--name brain-tasks-cluster-2 \

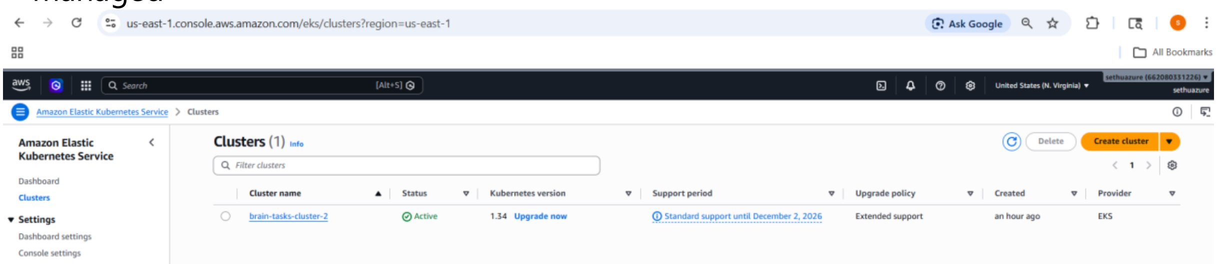
--region us-east-1 \

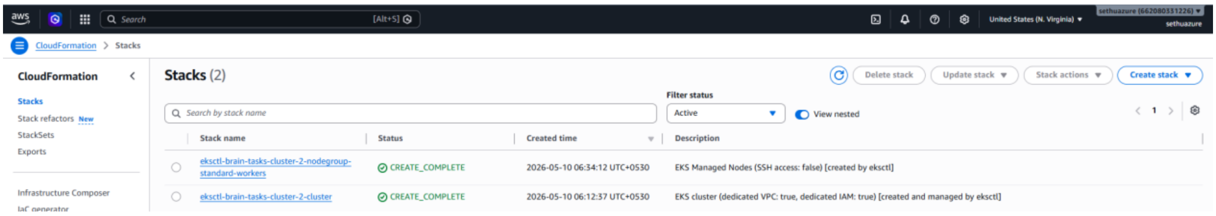
--nodegroup-name standard-workers \

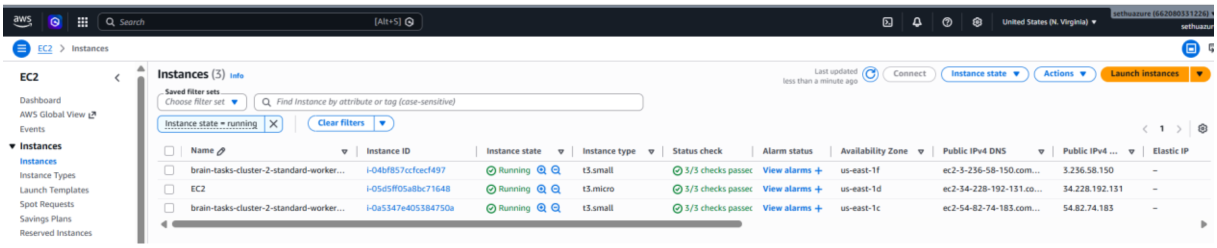
--node-type t3.small \

--nodes 2 \

--managed







Connect kubectl to cluster

aws eks update-kubeconfig \

--region us-east-1 \

--name brain-tasks-cluster-2

Verify nodes

kubectl get nodes

```
[ec2-user@ip-172-31-42-225 ~]$ kubectl get pods -n kube-system
NAME                      READY   STATUS    RESTARTS   AGE
aws-node-cz5q2            2/2     Running   0          2m32s
aws-node-z89ph            2/2     Running   0          2m32s
coredns-566b9b9d-gl2g5    1/1     Running   0          2m3s
coredns-566b9b9d-jxnmz    1/1     Running   0          2m3s
kube-proxy-tvf48          1/1     Running   0          116s
kube-proxy-tzlrx          1/1     Running   0          116s
[ec2-user@ip-172-31-42-225 ~]$ kubectl get nodes
NAME                          STATUS   ROLES    AGE   VERSION
ip-192-168-26-203.ec2.internal   Ready    <none>   28m   v1.34.7-eks-4136f65
ip-192-168-46-131.ec2.internal   Ready    <none>   28m   v1.34.7-eks-4136f65
[ec2-user@ip-172-31-42-225 ~]$
```

📄 STEP 7: Deploy to Kubernetes

Vi deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
      - name: brain-tasks
        image: 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
        ports:
        - containerPort: 3000
```

Vi service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks
  ports:
    - port: 80
      targetPort: 3000
~
```

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

```
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi deployment.yaml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi service.yaml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ kubectl apply -f deployment.yaml
deployment.apps/brain-tasks-deployment created
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ kubectl apply -f service.yaml
service/brain-tasks-service created
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ kubectl get svc
NAME                  TYPE           CLUSTER-IP      EXTERNAL-IP                                                               PORT(S)        AGE
brain-tasks-service   LoadBalancer   10.100.83.115   ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com   80:32221/TCP   13s
kubernetes            ClusterIP      10.100.0.1      <none>                                                                    443/TCP        52m
```

👉 Get URL:

kubectl get svc

ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com

8. Version Control (GitHub)

🚀 STEP 1: Create GitHub Repository

Go to 👉 GitHub

- Click **New Repository**
- Fill details:

- **Repository name**: brain-tasks-devops-new
- **Visibility**: Public (recommended for easy AWS access)
- ✅ Add README (optional)

- Click **Create Repository**

🚀 STEP 2: Clone Repo to Your System

git init

git clone https://github.com/sethuraman98/brain-tasks-devops-new.git

cd brain-tasks-devops

```
Last login: Sun May 10 01:24:54 2026 from 27.5.74.173
[ec2-user@ip-172-31-42-225 ~]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ec2-user/.git/
[ec2-user@ip-172-31-42-225 ~]$ git clone https://github.com/sethuraman98/brain-tasks-devops-new.git
Cloning into 'brain-tasks-devops-new'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
[ec2-user@ip-172-31-42-225 ~]$ cd Brain-Tasks-App/
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ ls
Dockerfile  README.md  deployment.yaml  dist  service.yaml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi buildspec.yml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ vi buildspec.yml
[ec2-user@ip-172-31-42-225 Brain-Tasks-App]$ ls
Dockerfile  README.md  buildspec.yml  deployment.yaml  dist  service.yaml
```

🚀 STEP 3: Create Required Files

touch Dockerfile buildspec.yml deployment.yaml service.yaml

We have to create buildspec.yml only remaining file already created and I just moved files from existing directory to new directory

```
version: 0.2

phases:
  install:
    commands:
      - echo Installing dependencies
      - aws --version

  pre_build:
    commands:
      - echo Logging into ECR
      - aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 662080331226.dkr.ecr.us-east-1.amazonaws.com

  build:
    commands:
      - echo Build started
      - docker build -t brain-tasks .
      - docker tag brain-tasks:latest 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks:latest

  post_build:
    commands:
      - echo Pushing image
      - docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks:latest
~
~
~
```

STEP 5: Clone your app inside this repo:

git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git app

Then move files:

mv app/* .

rm -rf app

🚀 STEP 6: Commit & Push

git add .

git commit -m "Final DevOps setup"

git push origin main

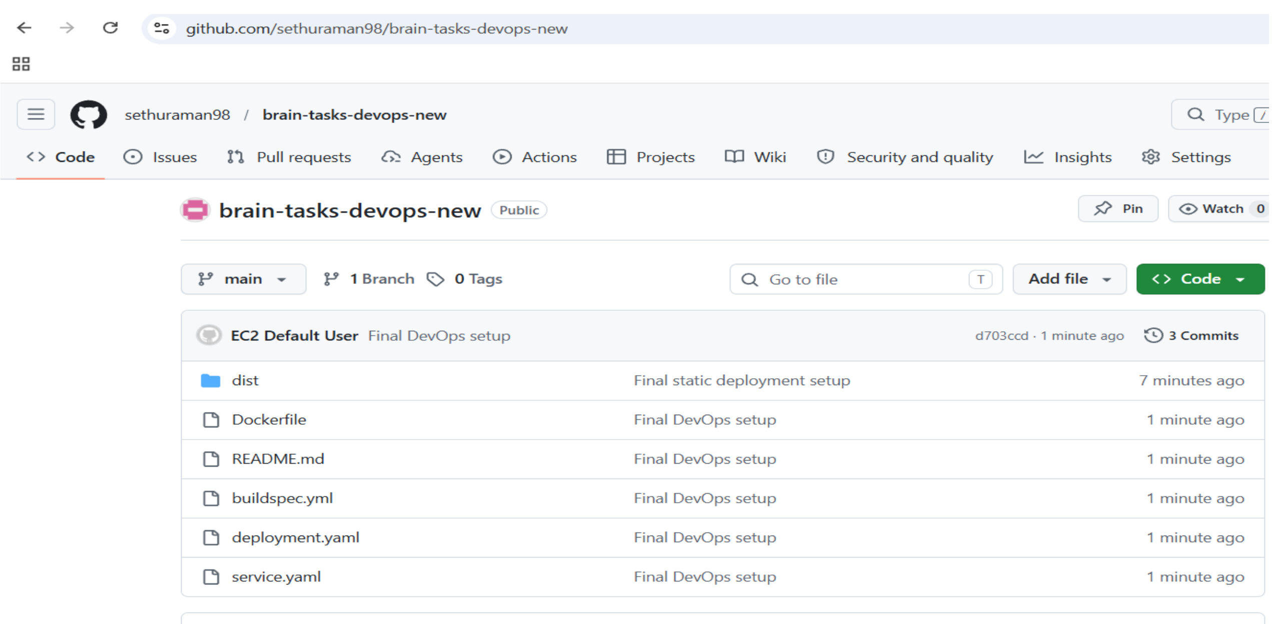
```
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ ls
Dockerfile  README.md  buildspec.yml  deployment.yaml  dist  service.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ git add .
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ git commit -m "Final DevOps setup"
[main d703ccd] Final DevOps setup
 Committer: EC2 Default User <ec2-user@ip-172-31-42-225.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 5 files changed, 149 insertions(+), 1 deletion(-)
 create mode 100644 Dockerfile
 create mode 100644 buildspec.yml
 create mode 100644 deployment.yaml
 create mode 100644 service.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ git push origin main
Username for 'https://github.com': sethuraman98
Password for 'https://sethuraman98@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 2.00 KiB | 2.00 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sethuraman98/brain-tasks-devops-new.git
   e099145..d703ccd  main -> main
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ |
```



git add .

git commit -m "Initial commit"

git push -u origin main

▭ 9. CodePipeline Setup

Pipeline Stages:

1. Source
   - GitHub repo
2. Build
   - AWS CodeBuild project
3. Deploy
   - Use CodeBuild or EKS deploy stage

👉 Ensure IAM roles allow:

- ECR access
- EKS access
- CloudWatch logs

Create CodeBuild Project

Go to:
👉 AWS CodeBuild

Create build project

Project Name : brain-tasks-build

Source Provider: GitHub

Repository: brain-tasks-devops-new

Branch: main

Environment Image: Managed image

Operating System: Amazon Linux

Runtime: Standard

Image: Latest standard image.

VERY IMPORTANT

Enable:

☑ Privileged

Without this:
✕ Docker build fails

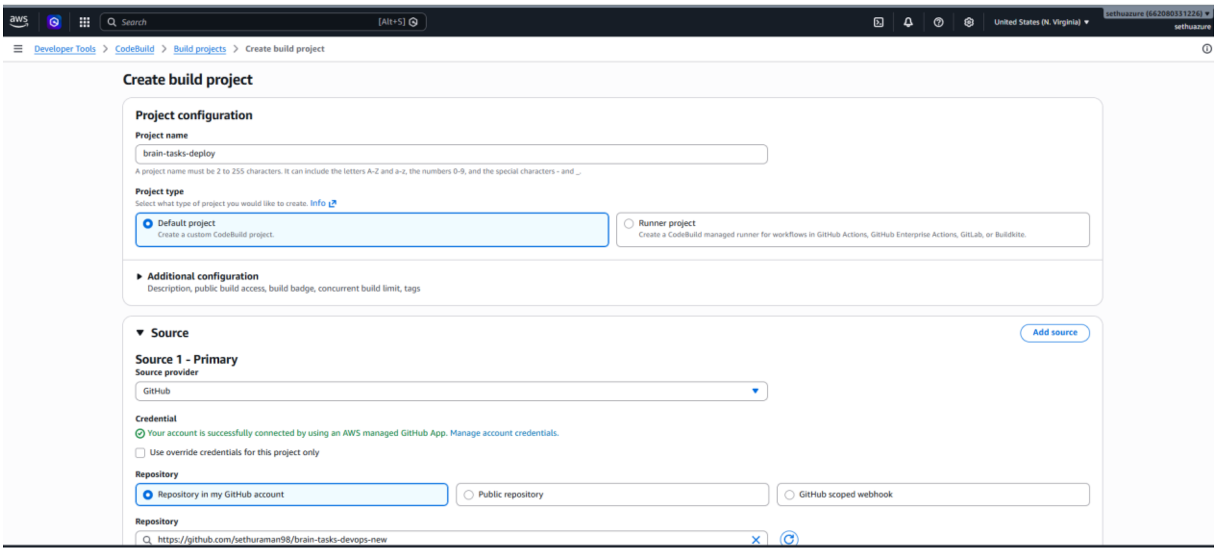
Service Role: New service role

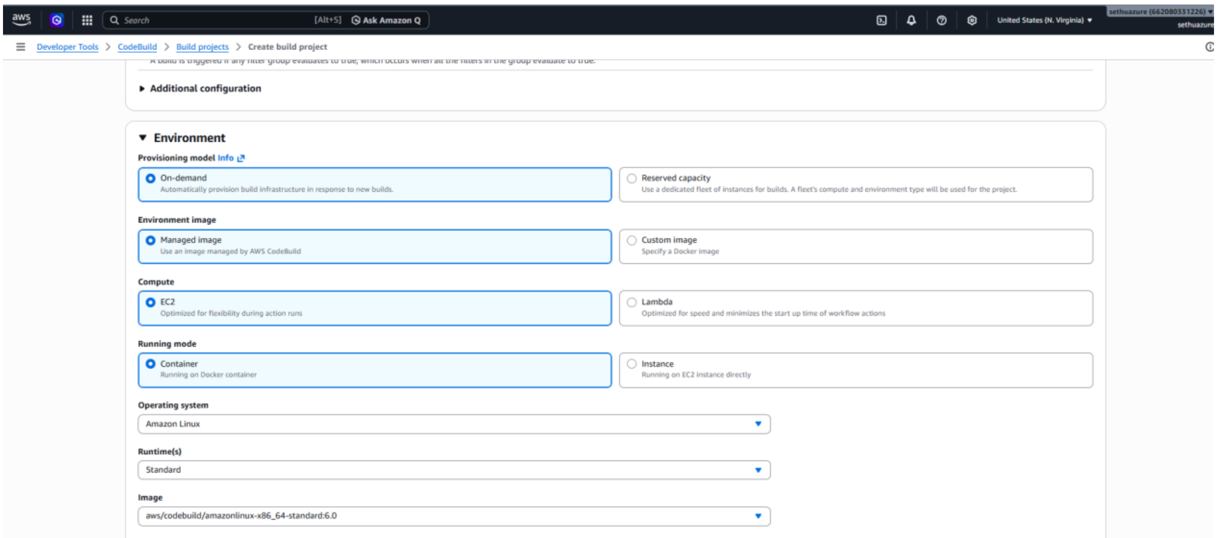
Role name: codebuild-brain-tasks-role

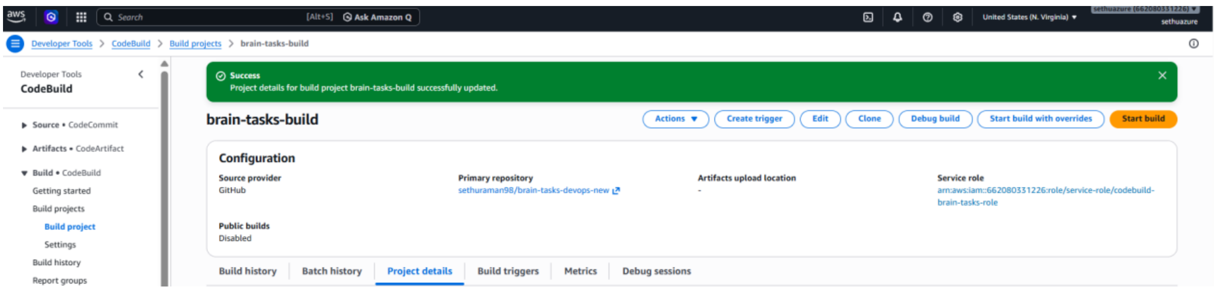
Buildspec: buildspec.yml

Artifacts: No artifacts

Logs: ☑ CloudWatch Logs







Attach IAM Policies

Go to:
👉 AWS Identity and Access Management → Roles

Search:

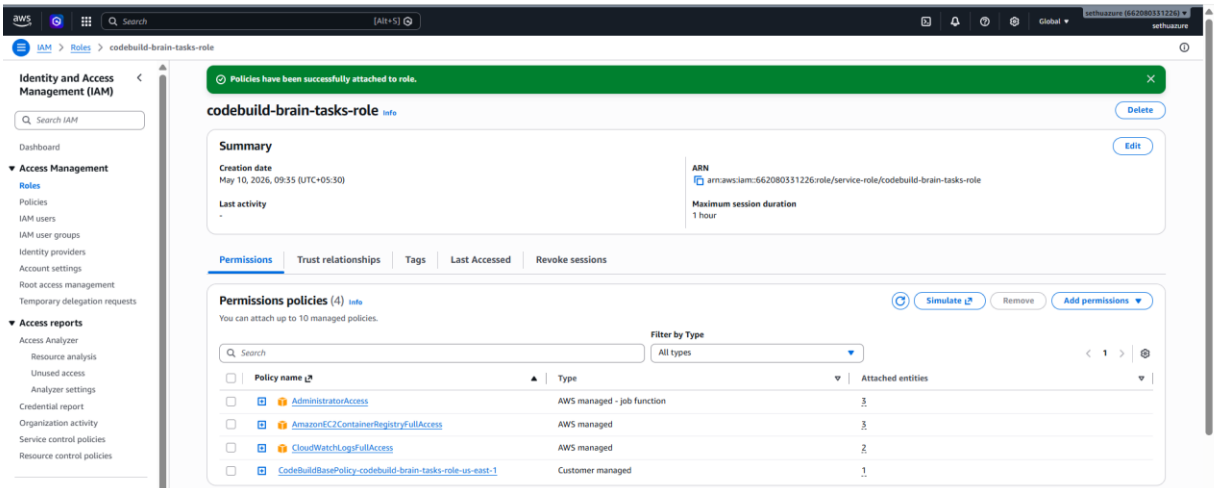
codebuild-brain-tasks-role

Attach:

AmazonEC2ContainerRegistryFullAccess

CloudWatchLogsFullAccess

AdministratorAccess

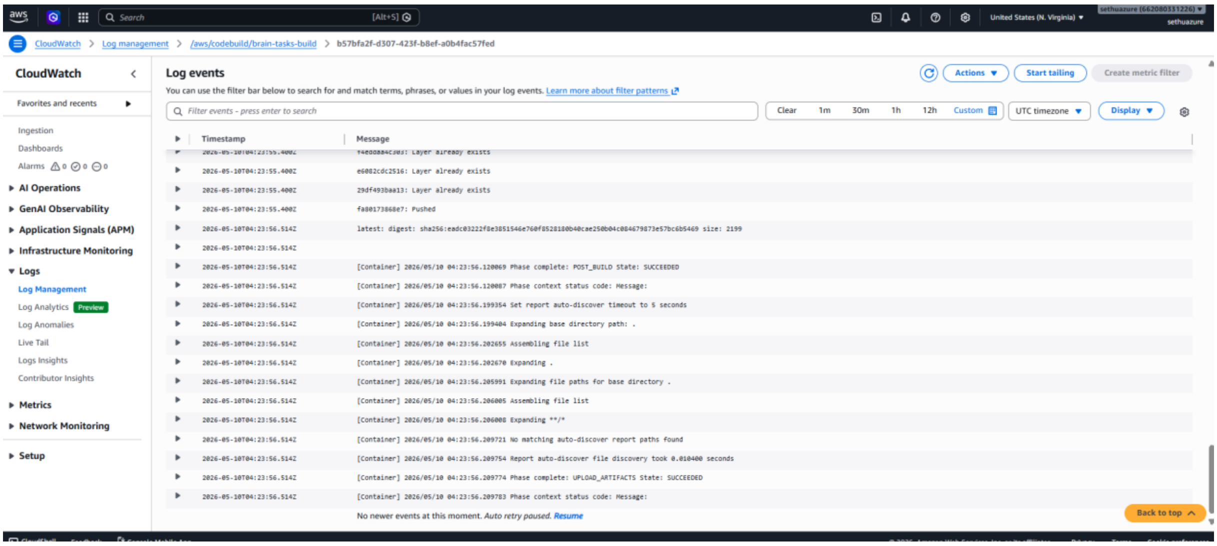


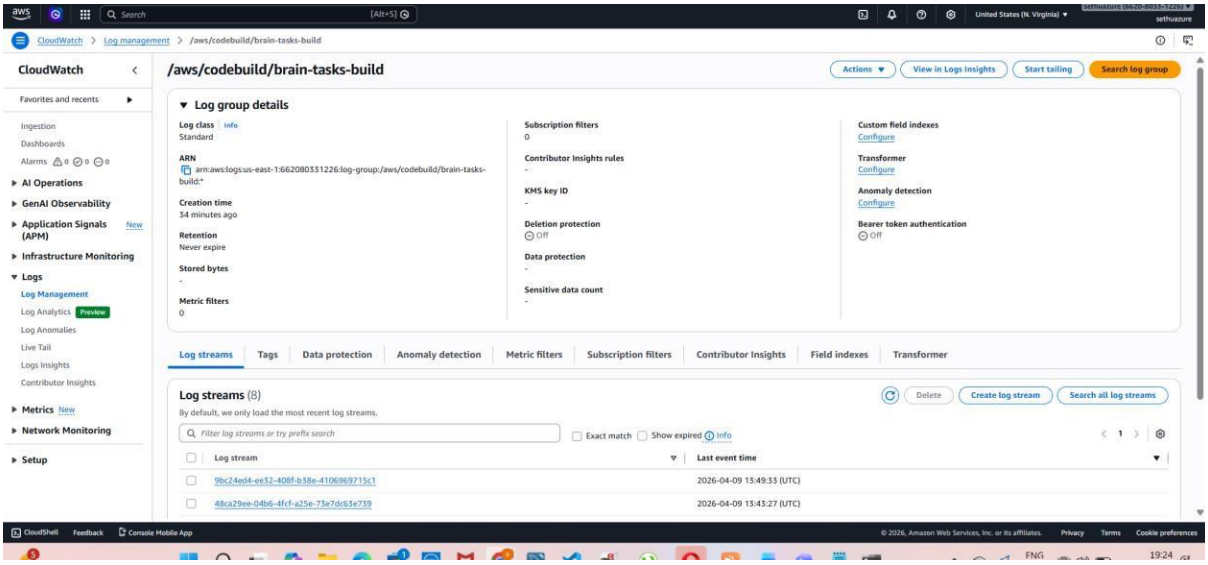
Start Build

Cloudwatch

CloudWatch Monitoring

CloudWatch Logs were enabled for AWS CodeBuild to monitor build execution, Docker image push activity, and deployment-related logs. The logs confirm successful completion of the build and post-build stages.

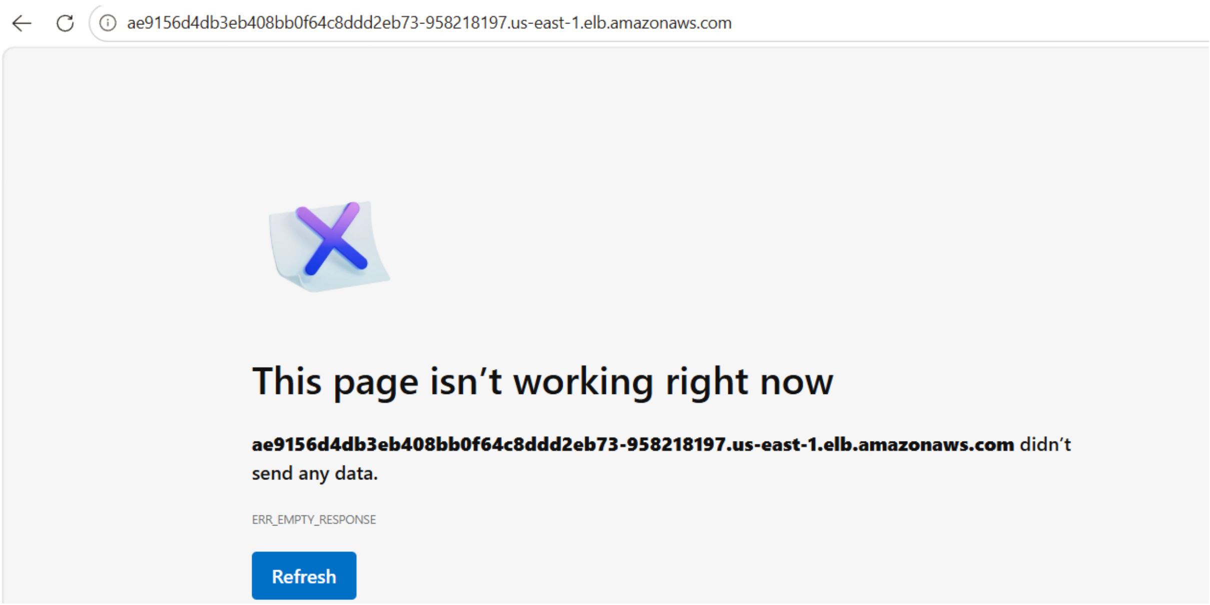




"I implemented a full DevOps CI/CD pipeline using AWS services including Docker, ECR, EKS, and CodeBuild to deploy a React application, which is accessible via a LoadBalancer URL."

Troubleshooting:

I didn't get the output. I have to check the ports.



Updated deployment.yaml file

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
      - name: brain-tasks
        image: 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
        ports:
        - containerPort: 80
```

Updated service.yaml file

```yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks
  ports:
    - port: 80
      targetPort: 80
~
```

Run:
kubectl apply -f deployment.yaml
kubectl apply -f service.yaml

Restart Deployment

kubectl rollout restart deployment brain-tasks-deployment

Verify Pods

kubectl get pods

kubectl get svc

```
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ vi deployment.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ vi service.yaml
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/brain-tasks-deployment configured
service/brain-tasks-service configured
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl rollout restart deployment brain-tasks-deployment
deployment.apps/brain-tasks-deployment restarted
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl get pods
NAME                                     READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-7746788b9-mbzqh   1/1     Running   0          8s
brain-tasks-deployment-7746788b9-p48j5   1/1     Running   0          7s
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ kubectl get svc
NAME                  TYPE           CLUSTER-IP      EXTERNAL-IP                                                               PORT(S)        AGE
brain-tasks-service   LoadBalancer   10.100.83.115   ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com   80:32221/TCP   3h20m
kubernetes            ClusterIP      10.100.0.1      <none>                                                                    443/TCP        4h13m
[ec2-user@ip-172-31-42-225 brain-tasks-devops-new]$ |
```

ApplicationLoadBalancerURL:

http://ae9156d4db3eb408bb0f64c8ddd2eb73-958218197.us-east-1.elb.amazonaws.com

Final Output: